

Recap from ENGR101

Ethical Thinking for Engineers

Part 2

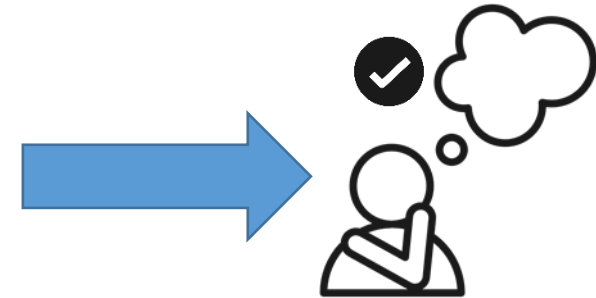
Ethical Theories – Deontology

Lecture by Enda Crossin

Our Goal: Applied Ethics

Use existing principles to make ethically sound decisions

1. Personal sense of right and wrong
- 2. Understanding of ethical theories**
3. Systematic decision-making process



What Makes an Action “Right” or “Wrong”?

- Ethical Theories Simplified

- **Rule-Based Thinking**
 - Do the “right” thing
- **Ends-Based Thinking**
 - Get the “best” outcome (consequences)
- **Virtues-Based Thinking**
 - Be a “good” person

Rule-Based* Thinking: Do the Right Thing

- An act is ethical if certain ethical rules are obeyed, regardless of the situation and potential outcomes
- Rule-based thinking is often absolutist – context does not matter

*aka deontology, duty-based thinking

Rule-Based Thinking: Pros and cons

- Strength: Simple to understand
 - If we follow the rules, we are ethical; if we break them, we are unethical
- Weakness: What are the rules? Who decides?
 - Obey “universal” moral laws (moral universalism)?
 - It is wrong to kill innocent people
 - It is wrong to steal
 - It is wrong to tell lies
 - It is right to keep promises
 - Obey obligations of faith / religion?
 - Obey the law?

Rule-Based Thinking: What are the Rules*?

* In this society, at this time

- Rule: Follow the law
- Rule: Treat others as you would wish to be treated (the Golden Rule)
- Rule: Follow your professional code of ethics
- Rule: Obey company/organisational policies
- Rule: People have fundamental **rights** that other people have a **duty** to respect
- Rule: Actions must be **just** and **fair**

Rule-Based Thinking: The Golden Rule

Rule: Treat others as you would wish to be treated

- "None of you [truly] believes until he wishes for his brother what he wishes for himself."
 - 40 Hadith Nawawi 13
- "Do to others as you would have them do to you"
 - Luke 6:31, Matthew 7:12
- "This is the sum of the Dharma [duty]: do naught unto others which would cause you pain if done to you."
 - Mahabharata 5:1517

- *Shortcomings*

- Assumes your needs and wants are the same as everyone else's
- Does not distinguish between competing stakeholders

Treat people the way *they* wish to be treated

Rule-Based Thinking: Professional Codes of Ethics

Rule: Follow your professional code of ethics.

- **All** engineering professional bodies in NZ and internationally have codes of ethics that all members **must** follow:
 - Engineering NZ Code of Ethical Conduct
 - IEEE (Institute of Electrical and Electronics Engineers) Code of Ethics
 - ACM/IEEE Software Engineering Code of Ethics
 - IET (Institution of Engineering and Technology) Rules of Conduct (UK)
 - Engineers Australia: Code of Ethics and Guidelines on Professional Conduct (AUS)
 - UK Royal Academy of Engineering: Statement of Ethical Principles
 - American Society of Civil Engineers: Code of Ethics
 - American Institute of Chemical Engineers: Code of Ethics
 - American Society of Mechanical Engineers: Code of Ethics

Rule-Based Thinking: Company Policy

Rule: Obey your organisation's code of ethics and other policies.

- A requirement of your employment. Some real examples.....:

- **WSP Opus**

- <https://www.wsp.com/en-US/who-we-are/corporate-responsibility/ethics-integrity>

- **Meridian Energy**

- <https://www.meridianenergy.co.nz/public/Investors/Governance/Policies/Code-of-Conduct.pdf>

- **Fisher and Paykel Healthcare**

- <https://www.fphcare.com/nz/our-company/sustainability/ethics-and-legal-compliance/>

- **Fulton Hogan**

- <https://www.fultonhogan.com/our-story/our-values-and-behaviours/>

Rule-Based Thinking: Rules about Rights

Rule: People have fundamental **rights** that other people have a **duty** to respect (legal requirements are the minimum)

- Examples are the right to:
 - Free consent
 - Privacy
 - Freedom of conscience
 - Due process
 - Be told the truth
 - Life and safety
 - Freedom of speech
 - Freedom of choice
- *Shortcomings*: It is not always possible to simultaneously uphold the rights of all people → conflicting rights

The Problem of Conflicting Rights

- Conflicting rights lead can lead to ethical dilemmas:
 - Right of one party to free speech vs right of another party to be told the truth
 - Right to privacy vs right to security (see next slide)
 - Some consider there to be a hierarchy of rights:
 - The right to have your own water supply vs the right of a council to require households to be grid-connected

Breaking Down Apple's iPhone Fight With the U.S. Government

By THE NEW YORK TIMES **UPDATED** March 21, 2016

What does the government want?

The F.B.I. has been trying to force Apple to help investigators gain access to an iPhone used by Syed Rizwan Farook in the December mass shooting in San Bernardino, Calif.

Apple vs. FBI Is Not About Privacy vs. Security — It's About How to Achieve Both

“Cyber security is a huge public safety concern. On the one hand, the FBI is confronted with its current dilemma of gathering evidence in a terrorism case. On the other hand, Apple is considering the strategic implications of a world in which strong encryption is ubiquitous but [not available] to consumers of American products.

There are no easy answers to this problem, and both sides are acting out of genuine interpretation of the law and their duties.” *Huffington Post*

Rule-Based Thinking: Rules about Justice

Rule: Actions must be just and fair (legal requirements are the minimum)

- Distributive justice (**equity**): Different treatment of people should not be based on arbitrary characteristics (discrimination, equal pay for equal work)
- Procedural justice (**impartiality**): Rules should be clearly stated and consistently and impartially enforced
- Compensatory justice (**fairness**)
 - Individuals should be compensated for injury by the party responsible
 - Individuals should not be held responsible for matters over which they have no control
- *Shortcomings*: It is not always possible to be equitable to all stakeholders

The NZ Bill of Rights Act 1990

<https://www.legislation.govt.nz/act/public/1990/0109/latest/DLM224792.html>

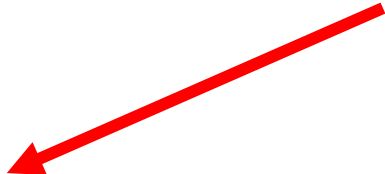
- The New Zealand Bill of Rights Act 1990 protects the civil and political rights of all New Zealanders.
- It is the legal “bottom line” for **rights and justice** in NZ.
- The Act covers the following categories of rights and freedoms:
 - Life and security of the person
 - Democratic and civil rights
 - Non-discrimination and minority rights
 - Search, arrest and detention
 - Criminal procedure
 - The right to justice

Rule-Based Thinking: UC Students

Rule: Follow the policies and codes of organisations to which you belong (for example the University of Canterbury)

Academic Integrity Policy

- Cheating
- Plagiarism
- Honesty in research and testing



<https://www.canterbury.ac.nz/about-uc/what-we-do/teaching/academic-integrity>

Student Code of Conduct

- Harassment and bullying
- General behaviour



<https://www.canterbury.ac.nz/about-uc/corporate-information/policies/student-code-of-conduct>

**Have you read with these two documents yet?
They outline what is acceptable behaviour on campus, as well as your rights and responsibilities as students of UC**

Rule-Based Thinking

- Weakness: What if the rules conflict with virtues and values? What if they conflict with each other? And what about the consequences?

Recap from ENGR101

Ethical Thinking for Engineers

Part 3

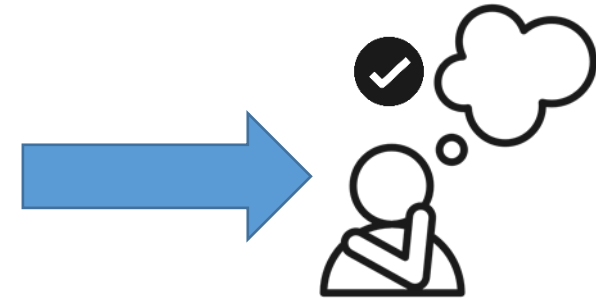
Ethical Theories – Consequentialism

Lecture by Enda Crossin

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Ends-Based* Thinking: Get the Best Outcome

- Weigh competing actions according to their consequences
 - Acts are ethical when they achieve the **best** outcome
 - “The ends justifies the means”
 - *Benefit*: Flexible, not absolute
 - *Shortcoming*: how to judge the “best” outcome?
 - Best for me?
 - Best for my family?
 - Best for most people?
- *aka teleology, consequentialism

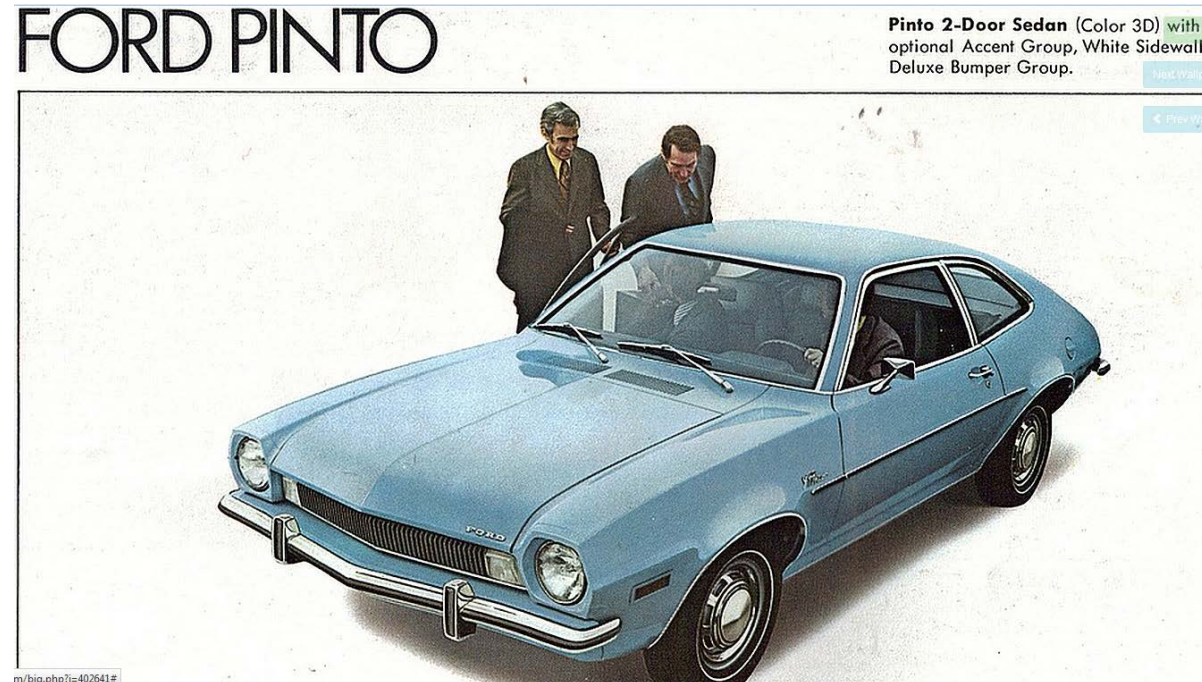
Ends-Based Ethics: Utilitarianism

- Acts are ethical when they achieve **the greatest good for the greatest number**
- *“It is the greatest happiness of the greatest number that is the measure of right and wrong”* - Jeremy Bentham, 1776
 - Assess good and bad outcomes
 - Consider the short and the long term
 - Consider the individual and all effected stakeholders
- Requires a **cost-benefit analysis**

Ends-Based Ethics: Utilitarianism

- *Shortcomings:*

- Only considers the overall happiness/suffering, not the distribution.
- Difficult to quantify “happiness” and “suffering”.



The Pinto Problem: Monetised Utilitarian Approach

- Two options:
 - 1) Redesign the fuel tank and retrofit all existing vehicles
 - 2) Leave as-is (do nothing)
- Apply a cost-benefit analysis
 - Determine the “cost” of each option – in this case, monetary
 - Determine the “benefit” of each option

Option 1: Redesign the fuel tank

- 12.5 million automobiles sold
- Fix cost per car: \$11 (at the time)
- Total cost: $\$11 \times 12.5 \text{ million} \rightarrow \137 million

Option 2: Ship the car as-is

- Death by burning for 180 buyers
- Serious burns for another 180 buyers
- 2,100 vehicles burned beyond all repair
- Value of Statistical Life (VSL, USA, 1972) = \$200,000 (approx. \$1.5 mill today)
- Cost of a serious burn = \$67,000
- Average resale = \$700 (lost if car destroyed)

The Pinto Problem: Monetised Utilitarian Approach

Redesign the gas tank

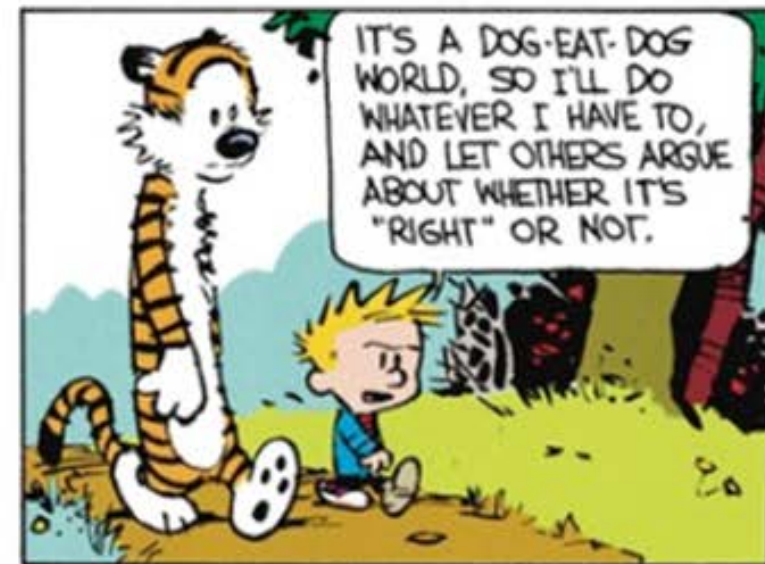
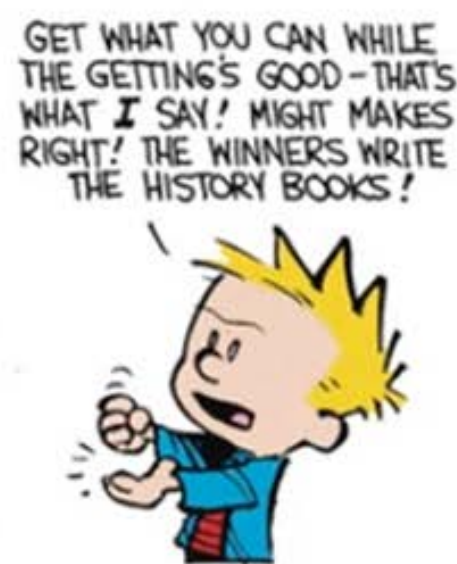
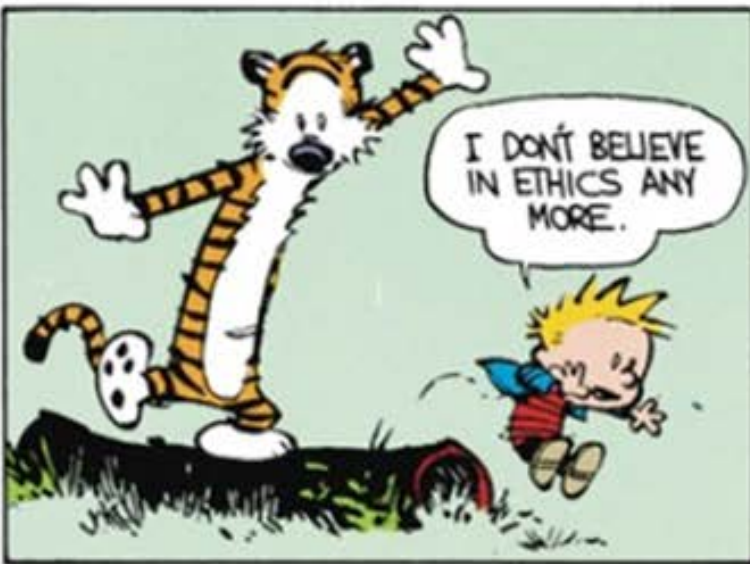
- 12.5 million automobiles sold
- Cost per car to implement redesign \$11

• Total cost: \$137 million

Ship the car as-is

- 180 deaths x \$200,000
- 180 injuries x \$67,000
- 2,100 burned out cars x \$700

• Total cost: \$49.5 million
(benefit as it was cheaper)



Best for whom?

Recap from ENGR101

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Part 4

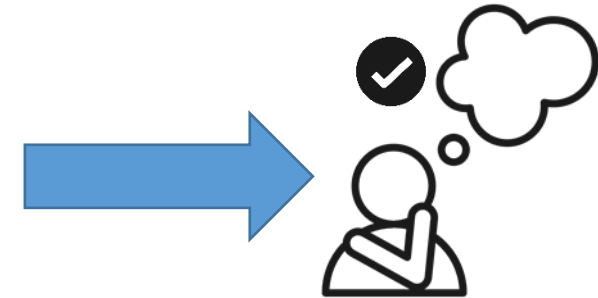
Ethical Theories – Virtue Ethics

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Virtues-Based Thinking: Be a Good Person

- Acts are ethical if they are what a “virtuous” person would do in the same circumstances.
 - A virtuous person would act with (for example) integrity, honesty, courage, compassion and self-regard. (ethical values)
 - Extremes are to be avoided however (Aristotle’s Golden Mean). For example, courage is a virtue, but too much is recklessness, and too little is cowardice
- *Shortcomings*: Assumes everyone values the same virtues and acts accordingly.

Virtues-Based Thinking: Some Virtues

- Honesty
- Courage
- Compassion
- Self-regard
- Generosity
- Tolerance
- Love
- Fidelity
- Integrity
- Fairness
- Self-control
- Prudence

Virtues-Based Thinking: Cultures and Biases

- What we consider as virtues depends on our **cultures** and **biases**
 - Need to consider what other cultures value
- We all have **cognitive biases**
 - We all create our own perception of reality
 - This perception affects how we behave and think

yourbias.is



anchoring

The first thing you judge influences your judgment of all that follows.

Human minds are associative in nature, so the order in which we receive information helps determine the course of our judgments and perceptions.

Be especially mindful of this bias during financial negotiations such as houses, cars, and salaries. The initial price offered is proven to have a significant effect.



sunk cost fallacy

You irrationally cling to things that have already cost you something.

When we've invested our time, money, or emotion into something, it hurts us to let it go. This aversion to pain can distort our better judgment and cause us to make unwise investments.

To regain objectivity, ask yourself: had I not already invested something, would I still do so now? What would I counsel a friend to do if they were in the same situation?



availability heuristic

Your judgments are influenced by what springs most easily to mind.

How recent, emotionally powerful, or unusual your memories are can make them seem more relevant. This, in turn, can cause you to apply them too readily.

Try to gain different perspectives and relevant statistical information rather than relying purely on first judgments and emotive influences.



curse of knowledge

Once you understand something you presume it to be obvious to everyone.

Things make sense once they make sense, so it can be hard to remember why they didn't. We build complex networks of understanding and forget how intricate the path to our available knowledge really is.

When teaching someone something new, go slow and explain like they're ten years old (without being patronizing). Repeat key points and facilitate active practice to help embed knowledge.



confirmation bias

You look for ways to justify your existing beliefs.

We are primed to see and agree with ideas that fit our preconceptions, and to ignore and dismiss information that conflicts with them.

Think of your ideas and beliefs as software you're actively trying to find problems with rather than things to be defended.

"The first principle is that you must not fool yourself -- and you are the easiest person to fool."
- Richard Feynman



dunning-kruger effect

The more you know, the less confident you're likely to be.

Because experts know just how much they don't know, they tend to underestimate their ability, but it's easy to be over-confident when you have only a simple idea of how things are.

"The whole problem with the world is that fools and fanatics are so certain of themselves, yet wiser people so full of doubts."
- Bertrand Russell



belief bias

If a conclusion supports your existing beliefs, you'll rationalize anything that supports it.

It's difficult for us to set aside our existing beliefs to consider the true merits of an argument. In practice this means that our ideas become impervious to criticism, and are perpetually reinforced.

A useful thing to ask is 'when and how did I get this belief?' We tend to automatically defend our ideas without ever really questioning them.



self-serving bias

You believe your failures are due to external factors, yet you're personally responsible for your successes.

Many of us enjoy unearned privileges, luck and advantages that others do not. It's easy to tell ourselves that we deserve these things, whilst blaming circumstances when things don't go our way.

When judging others, be mindful of how this bias interacts with the just-world hypothesis, fundamental attribution error, and the in-group bias.



backfire effect

When your core beliefs are challenged, it can cause you to believe even more strongly.

We can experience being wrong about some ideas as an attack upon our very selves, or on tribal identity. This can lead to motivated reasoning which causes us to double-down, despite disconfirming evidence.

"It ain't what you don't know that gets you into trouble. It's what you know for sure that just ain't so."
- Mark Twain



barnum effect

You see personal specifics in vague statements by filling in the gaps.

Because our minds are given to making connections, it's easy for us to take nebulous statements and find ways to interpret them so that they seem specific and personal.

Psychics, astrologers and others use this bias to make it seem like they're telling you something relevant. Consider how things might be interpreted to apply to anyone, not just you.



groupthink

You let the social dynamics of a group situation override the best outcomes.

Descent can be uncomfortable and dangerous to one's social standing, and so often the most confident or first voice will determine group decisions.

Rather than openly contradicting others, seek to facilitate objective means of evaluation and critical thinking practices as a group activity.



negativity bias

You allow negative things to disproportionately influence your thinking.

The pain of loss and hurt are felt more keenly and persistently than the fleeting gratification of pleasant things. We are primed for survival, and our aversion to pain can distort our judgment for a modern world.

Pro-and-con lists, as well as thinking in terms of probabilities, can help you evaluate things more objectively than relying on a cognitive impression.



declinism

You remember the past as better than it was, and expect the future to be worse than it will likely be.

Despite living in the most peaceful and prosperous time in history, many people believe things are getting worse. The 24 hour news cycle, with its reporting of overtly negative and violent events, may account for some of this effect.

Instead of relying on nostalgic impressions of how great things used to be, use measurable metrics such as life expectancy, levels of crime and violence, and prosperity statistics.



framing effect

You allow yourself to be unduly influenced by context and delivery.

We all like to think that we think independently, but the truth is that all of us are, in fact, influenced by delivery, framing and subtle cues. This is why the ad industry is a thing, despite almost everyone believing they're not affected by advertising messages.

Only when we have the intellectual humility to accept the fact that we can be manipulated, can we hope to limit how much we are. Try to be mindful of how things are being put to you.



fundamental attribution error

You judge others on their character, but yourself on the situation.

If you haven't had a good night's sleep, you know why you're being a bit slow, but if you observe someone else being slow you don't have such knowledge and so might presume them to just be a slow person.

It's not only kind to view others' situations with charity, it's more objective too. Be mindful to also err on the side of taking personal responsibility rather than justifying and blaming.



halo effect

How much you like someone, or how attractive they are, influences your other judgments of them.

Our judgments are associative and automatic, and so if we want to be objective we need to consciously control for irrelevant influences. This is especially important in a professional setting.

If you notice that you're giving consistently high or low marks across the board, it's worth considering that your judgment may be suffering from the halo effect.



optimism bias

You overestimate the likelihood of positive outcomes.

There can be benefits to a positive attitude, but it's unwise to allow such an attitude to adversely affect our ability to make rational judgments (they're not mutually exclusive).

If you make rational, realistic judgments you'll have a lot more to feel positive about.



pessimism bias

You overestimate the likelihood of negative outcomes.

Pessimism is often a defense mechanism against disappointment, or it can be the result of depression and anxiety disorders.

Perhaps the worst aspect of pessimism is that even if something good happens, you'll probably feel pessimistic about it anyway.



just world hypothesis

Your preference for a just world makes you presume that it exists.

A world in which people don't always get what they deserve, hard work doesn't always pay off, and injustice happens is an uncomfortable one that threatens our preferred narrative. However, it is also the reality.

A more just world requires understanding rather than blame. Remember that everyone has their own life story; we're all fallible, and bad things happen to good people.



in-group bias

You unfairly favor those who belong to your group.

We presume that we're fair and impartial, but the truth is that we automatically favor those who are most like us, or belong to our groups.

Try to imagine yourself in the position of those in out-groups; whilst also attempting to be dispassionate when judging those who belong to your in-groups.



placebo effect

If you believe you're taking medicine it can sometimes 'work' even if it's fake.

The placebo effect can work for stuff that our mind influences (such as pain) but not so much for things like viruses or broken bones.

Homeopathy, acupuncture, and many other forms of natural 'medicine' have been proven to be no more effective than placebo. Keep a healthy body and bank balance by using evidence-based medicine from a qualified doctor.



bystander effect

You presume someone else is going to do something in an emergency situation.

When something terrible is happening in a public setting we can experience a kind of shock and mental paralysis that distracts us from a sense of personal responsibility. The problem is that everyone can experience this sense of deindividuation in a crowd.

If there's an emergency situation, presume to be the one who will help or call for help. Be the change you want to see in the world.



reactance

You'd rather do the opposite of what someone is trying to make you do.

When we feel our liberty is being constrained, our inclination is to resist, however in doing so we can over-compensate.

Be careful not to lose objectivity when someone is being coercive/manipulative, or trying to force you to do something. Wisdom springs from reflection, fully from reaction.



spotlight effect

You overestimate how much people notice how you look and act.

Most people are much more concerned about themselves than they are about you. Absent overt prejudices, people generally want to like and get along with you as it gives them validation too. Instead of worrying about how you're being judged, consider how you make others feel. They'll remember this much more, and you'll make the world a better place.



24 cognitive biases stuffing up your thinking

What does NZ culture and identity mean?

Cultural norms:

- “She’ll be right”
 - Everything will be OK later, so it doesn’t matter what action we take now
- Number 8 fencing wire
 - We can get by with a simple solution and minimal input , even if it’s not quite right
- Tall poppies
 - A propensity to criticise people who are succeeding, or who are proud of their success
- Compassion (kindness)
- Tikanga Māori; respect and acknowledgement of cultural protocols

ENGR101

Ethical Thinking for Engineers

Part 5

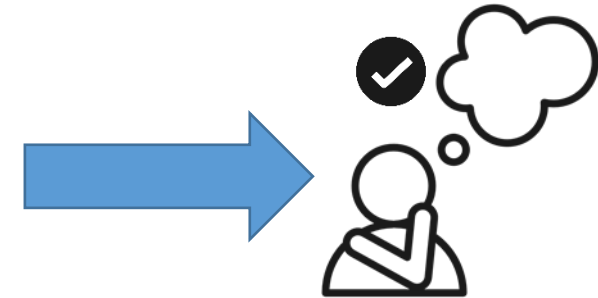
Applied Ethics

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Ethics: Beyond the letter of the law

- Lowest standard (may be legal, but not ethical)
- The letter or the spirit of the law?



What Makes an Action Right or Wrong: Ethical Theories Simplified

- **Rule** Based Ethics (deontology, duty ethics)
 - The belief that an act can be judged ethical if it is the right thing to do, regardless of the consequences
 - Do the right thing by following rules
- **Ends** Based Ethics (teleology, consequentialism)
 - The belief that an act can be judged ethical or unethical depending on its **consequences**.
 - Get the best outcome (the ends justifies the means)
- **Virtues** Based Ethics
 - The belief that an act can be judged ethical if the intent of the doer is to act as a virtuous person would
 - Be a good person

Blending Ethical Theories

- People do not have to align themselves with a single theory
 - Do not have to choose between being a strict deontologist or a strict teleologist
 - Rule Consequentialism
 - Follow the rules, but when they conflict, choose based on the consequences
 - Shortcoming: as with utilitarianism, what happens to the “lesser number”?
- Use the theories as tools to make ethical decisions you can defend.

Blending Ethical Theories: Weinstein's Five Ethical Principles

1. Do no harm
 - Prevent harm
 - “All that is necessary for evil to flourish is for good to do nothing” *Edmund Burke (attrib.)*
2. Make things better
 - Do good
3. Respect others
4. Be fair
5. Be compassionate

A mixture of
consequences, rules
and virtues

Engineering NZ Code of Ethical Conduct



What is the purpose of the Engineering NZ Code of Ethical Conduct?

A dark blue poster with white text. At the top, it says 'STAND UP FOR WHAT'S RIGHT' in large, bold, sans-serif capital letters, with 'Code of Ethical Conduct' in a smaller font below it. To the right of the title is a white logo consisting of two interlocking loops. Below the title is a numbered list of eight principles. At the bottom left is the website 'www.engineeringnz.org' and at the bottom right is the Engineering New Zealand logo, which includes the text 'engineering new zealand' and 'the authority for engineering excellence' below it.

**STAND UP FOR
WHAT'S RIGHT**
Code of Ethical Conduct

- 1 Protect people's health and safety
- 2 Consider the environment
- 3 Report adverse consequences
- 4 Act competently
- 5 Behave appropriately
- 6 Warn about ignoring advice
- 7 Maintain confidentiality
- 8 Report significant breaches

www.engineeringnz.org

engineering
new zealand
the authority for engineering excellence

“The Code of Ethical Conduct sets out the shared expectations of ethical practice that engineers have agreed to uphold. It’s a benchmark that confirms the expected standards of conduct for members....

.... The Code of Ethical Conduct is designed to help you make an ethical decision, and the decision that is best aligned to the standards of ethical behaviour set by your profession.”

Engineering NZ Code of Ethical Conduct

5. Behave appropriately

In performing, or in connection with, your **engineering activities** you

a. must

- i. act with **honesty, objectivity, and integrity**; and
- ii. treat people with **respect** and **courtesy**; and
- iii. disclose and appropriately manage conflicts of interest; and...

Obligations relating to personal conduct

5. Behave appropriately

In performing, or in connection with, your **engineering activities** you

b. must not

- i. offer or promise to give to any person anything intended to improperly influence a decision relating to your engineering activities; or
- ii. accept from any person anything intended to improperly influence your engineering activities; or
- iii. otherwise engage in, or support, corrupt practices.

Summary of Professional Engineering Codes

- Engineers do not lie, cheat or steal
- Engineers treat people fairly, impartially and with respect
- Engineers always have the best interests of the public and the environment in mind.
- Engineers have a responsibility to clients and colleagues to behave in a professional manner and do their best work within their area of competence only.

UC Engineering Student Code of Code of Conduct

- I will:

- **Respect others and their rights**, on campus and online. **Discrimination, harassment and bullying are unacceptable in any form.**
- Respect the property of others.
- Respect the environment.
- Follow New Zealand law and University of Canterbury policies.
- Behave in a manner appropriate to a professional engineering student in NZ. This includes a mature attitude to dress, choice of language, substance use and punctuality.
- Be aware of the effects of my behaviour on others.
- Speak out against a breach of the code

- I will not:

- Cheat
- Plagiarise
- Collude
- Falsify
- Fail to participate fully in group work or
- Otherwise attempt to get a better grade than I have earned.

**Compiled by ENGR 101 students
from 2017-19 to provide a simple
summary of all relevant UC
Student Codes and Policies**

UC Academic Integrity Module

UC AKO | LEARN My Courses English (en)

Academic Integrity Module – Semester One 2022

Dashboard / My courses / Academic Integrity Module – Semester One 2022

Start here!



Nau mai, tauti mai, welcome to the Te Whare Wānanga o Waitaha
University of Canterbury (UC) **Academic Integrity Module**

Participants

Badges

Grades

Show all sections

Sections

- Start here!
- Aromatawai | Assessment
- 1: Respect for ideas
- 2: Working with academic integrity and getting support
- 3: Where the line gets blurred
- 4: Processes at UC
- 5: Resources and FAQ

- Academic integrity involves **courage, trust, honesty, fairness, responsibility** and **respect**. Acting with academic integrity is key to success at UC as well as to the value, use and reputation of your degree beyond UC.

Some simple ethical tests

- Harms test
 - Does this option do less harm than any alternative (utilitarianism)
- Publicity test
 - Would I be happy defending this option in the media?
- Professional colleague test
 - What would my colleagues say if I described the problem and suggested this option as my solution? (rules)
- Reversibility test
 - Would I still choose this option if I was one of those negatively affected by it? (golden rule)
- Virtue test
 - Would a good person choose this option?

In other words:

- Is the action legal?
- Is it right?
- Who will be affected, and how?
- Does it fit with the values and ethics of my company/my profession/me?
- How will I feel afterwards?
- Will it reflect poorly on the company/my profession/me?

The Eight Step Method of Ethical Decision Making

Adapted from Langenderfer and Rockness

STEPS	Descriptive criteria
1. Identify the facts	Define the scope of the problem
2. Identify the ethical issues and the stakeholders involved	List the significant stakeholders and the way in which they could be affected
3. Identify the ethical principles, rules and values related to the situation	Consider how these guidelines influence individuals, the company, the profession and society in general
4. Identify possible courses of action	List the alternatives of what can and cannot be done
5. Apply principles, rules and values to each possible course of action	Determine whether there is a clear course of action. Discard those that are unacceptable.
6. Evaluate the consequences of each remaining course of action	Identify the possible outcomes of alternatives (both positive and negative). Consider impact vs likelihood.
7. Discuss alternatives/ take advice	Talking with a trusted person can help gain greater perspective regarding alternatives
8. Make a decision	If you can't, repeat steps 4 - 7

Example scenario:

You are considering specifying the use of lithium-ion batteries in a design.

1. Identify the Facts: Lithium-Ion Batteries

- ✓ More efficient, lighter and smaller than lead-acid batteries
- ✓ Widely used and essential for smart phones and laptops (convenience)
- ✓ Help with transition to renewable energy (inc. greenhouse gas reduction)



"This is where your smartphone battery begins", Washington Post, 2016

Classification: In-Confidence

1. Identify the Facts: Lithium-Ion Batteries

- ✓ More efficient, lighter and smaller than lead-acid batteries
- ✓ Widely used and essential for smart phones and laptops (convenience)
- ✓ Help with transition to renewable energy (inc. greenhouse gas reduction)
- x Needs cobalt (human cost of mining)
- x Needs lithium (loss of pristine environments)
- x Needs graphite (pollution)

2. Identify the ethical issues and the stakeholders

- The stakeholders:
 - The exploited workers – thousands
 - The companies making and using LiBs (and their shareholders and employees) - thousands
 - The users of the product - millions
 - People affected by mining pollution - thousands
 - People affected by climate change – everyone
- The issues
 - Health and safety of workers, exploitation, pollution

3. Identify the relevant principles, rules and values

- Rules:
 - Is it legal?
 - Engineering code of ethics
 - Health and safety of people
 - Consider the environment
 - Respect people
 - Act appropriately
 - Company policies
 - Rights of people and duty of protection?
- Virtues/ Values:
 - Increase shareholder wealth (profit)?
 - Weinstein
 - Do no harm
 - Do good

4. Identify alternative courses of action

- Option 1: Business as usual (specify the batteries)
- Option 2: Stop making the product until a “better” battery is developed
- Option 3: Use only batteries made from recycled materials (no new raw materials)
- Option 4: Use only raw materials that are certified “ethically sourced”

5. Apply principles, rules and duties identified in 3.

		Legal?	Eng Code	Company Policy	Rights	Do no harm	Do good?	Maximise profit
Actions	1: Specify LIBs	✓			X			✓
	2: Stop manufacture	✓			✓			X
	3: Use only recycled materials	✓			✓			X
	4: Use only ethically sourced materials	✓			✓			X

6. Evaluate the consequences

- What is the greatest good for the greatest number?

	Option	Workers	Companies	Product Users	People effected by mining pollution	People effected by climate change
Action	1: Specify LIBs					
	2: Stop manufacture					
	3: Use only recycled materials					
	4: Use only ethically sourced raw materials					

6. Evaluate the consequences

- What is the greatest good for the greatest number?

	Option	Workers	Companies	Product Users	People effected by mining pollution	People effected by climate change
Action	1: Specify LIBs			Good for users (millions each get a cheaper product)		
	2: Stop manufacture			Bad for users (millions don't get product)		
	3: Use only recycled materials			Less bad for users (product is more expensive)		
	4: Use only ethically sourced raw materials			Less bad for users (product is more expensive)		

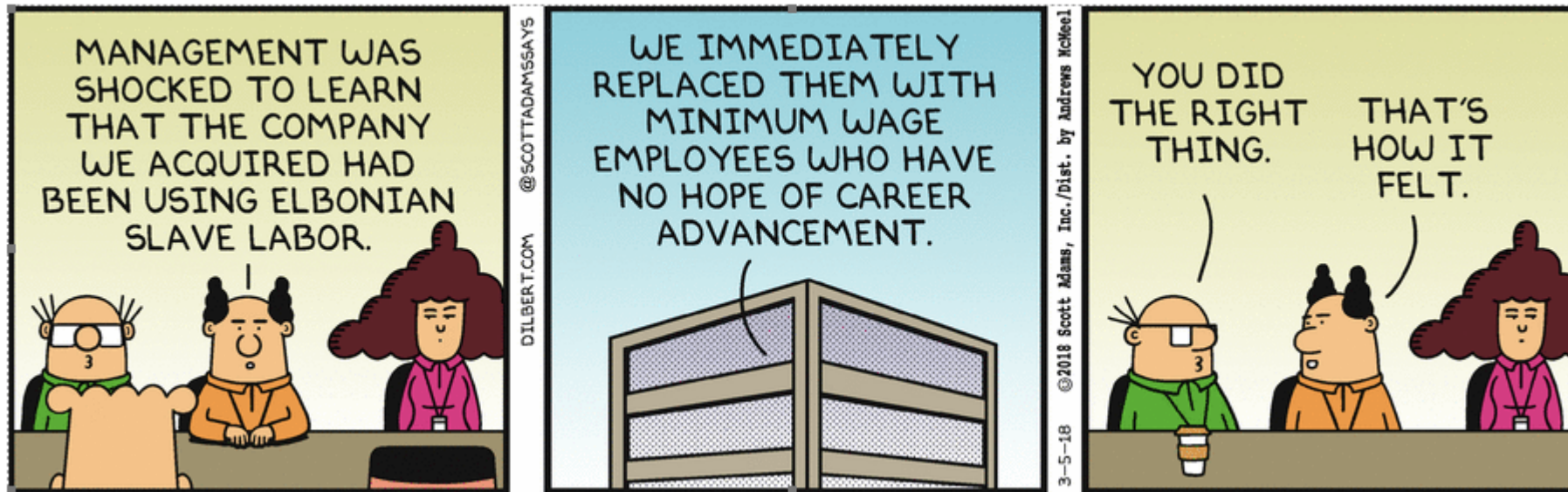
7. Discuss with others

.... ??????

8. Make a decision

.... ??????

Ethics are not binary



IF YOU DON'T LIKE YOUR OPTIONS, MAKE SOME MORE!

Most engineers see engineering as a profession that serves the public interest.

We WANT to behave ethically.

But what does that *look* like?

New engineer spotted alleged defects in Christchurch high-rise from the street

Jun 06 2019



JOHN KIRK-ANDERSON/STUFF

Concerns have been raised about the building at 230 High St, the seven-level building with blue-tinted glass pictured centre, in central Christchurch.

DISCIPLINARY COMMITTEE DECISION REGARDING A COMPLAINT ABOUT JOO CHO

Confidential to parties

In accordance with:

Engineering New Zealand Rules 2019
Engineering New Zealand Disciplinary Regulations 2017
Chartered Professional Engineers of New Zealand Act 2002
Chartered Professional Engineers of New Zealand Rules (No 2) 2002

Prepared by

Jenny Culliford FEngNZ (Ret.)

Chair of the Disciplinary Committee

Peter Boardman FEngNZ CPEng IntPE(NZ)
Rachel Wright CMEngNZ CPEng IntPE(NZ)
David Naulls, nominated by Consumer New Zealand
Anita Killeen, Barrister of the High Court of New Zealand
Members of the Disciplinary Committee

21 December 2021



“In closing, we wish to acknowledge the assistance of the complainant and his firm in this matter. Considerable time and effort has been spent by these engineers in acting on their concerns about the building at 230 High Street.”

If an engineer has reasonable grounds to believe that an engineering matter has, or could have, adverse consequences, the Code of Ethical Conduct requires them to bring the matter to the notice of the relevant regulatory body. The Code also requires engineers to take reasonable steps to safeguard the health and safety of people. We thank the complainant for their commitment to ethical practice.”